PEARLBOUND

GAME DESIGN DOCUMENT / MOBILE KICKOFF / ASSIGNMENT 8-9

01 / TITLE + PITCH

Pearlbound is a 60-second seabed expedition: guide a diver through ancient underwater ruins, gather pearls and dodge sinking mines until the rescue beacon activates.

02 / GENRE + PLATFORM

Solo arcade survival platformer for Android, landscape only. Target session: one minute, followed by a replay. Unity 6 and C#. iPhone and iPad profiles are also used to test the mobile UI layout.

03 / DIMENSION

2D side view, pixel art and sprite animation. The diver stays near the seabed and kicks upward from the floor or a ledge. The camera shows one arena. This MVP uses the familiar grounded-jump physics, rather than unrestricted swimming.

04 / CORE LOOP

MOVE -> DODGE -> COLLECT -> SURVIVE / REPLAY.
Traverse the seabed and two reef ledges, avoid mines, gather three pearls and collect healing air bubbles. Three starting hearts and two seconds of invulnerability after damage make mistakes recoverable. The rescue arch lights up after 60 seconds; death restarts the run.

05 / CONTROLS

Bottom-left LEFT / RIGHT: hold to move. Bottom-right KICK: one upward impulse while grounded; BUBBLE: hold to emit a bubble every 0.3 seconds. Bubbles are visual projectiles in this kickoff, with no damage mechanic. Top-right PAUSE. Menu PLAY / QUIT; pause RESUME / RESTART / MENU. Keyboard fallback: A/D or arrows, Space, X. Touch targets: at least 120 x 120 reference pixels.

06 / CONTENT

One seabed arena; one animated diver (Idle, swimming movement, upward kick); two reef ledges; three pearl pickups; one rescue arch. Three spawned types: small mine (-1 HP), large mine (-2 HP, delayed burst) and healing air bubble (+1 HP). One patrolling fish is planned for week 10, not present in the kickoff. No additional levels.

SCOPE & DELIVERY

ONE UNDERWATER WORLD, ONE FINISHABLE MOBILE LOOP

07 / ART + SOUND

Existing licensed pixel art: Underwater Diving Pack by Luis Zuno / ansimuz (CC0 / public domain) for the diver, mines, ruins, seabed and bubbles; 16x16 Pearl by MallardTV (CC0) for pearls. Sources: opengameart.org/content/underwater-diving-pack and /content/16x16-pearl. Original notices are retained. Deep blue water, sea-green ruins and pale readable UI form one visual world. Kenney Digital Audio (CC0) supplies the existing shoot, hurt and pickup sounds; the art pack music is not used.

08 / MVP FOR THE WEEK-15 BUILD

A complete menu -> seabed run -> pause / resume / restart / menu loop. Responsive hearts, pearl score and dive timer; touch controls; notch-safe layout. Keep spawning, healing, damage, shot cooldown and the 60-second rescue beacon. Later assignments add one fish enemy, local best-score/settings saving and audio/effect polish. Individual work until week 14.

09 / CUT LIST

No unrestricted vertical swimming, multiplayer, accounts, online leaderboard, shop, ads, inventory, upgrades, procedural maps or extra worlds. No networking or backend. Finish the single arena and mobile UI before expanding content.

10 / PLAN: WEEKS 9-13 + ROLES

Week	Reviewable result
9	GDD, licensed underwater art, menu/HUD/pause, safe area, mobile controls and Git feature PR.
10	Add one patrolling fish using the enemy-AI lesson; balance mines and health pickups.
11	Save the local best pearl score and sound preference; verify reopening the game.
12	Refine sound levels, bubble bursts and clear damage feedback.
13	Freeze features; test Android and the three layouts, fix issues and rehearse the English walkthrough.

Roles: solo: game design, implementation, asset integration, testing and presentation. No team roles before week 14.

Kickoff evidence: HUD on 16:9 phone, notched phone and 4:3 tablet; menu and pause screenshots; 3+ meaningful commits and a merged feature PR. Submission includes a 2-3 minute English MP4 with the student's own voice.